

A02 Technical Report: Hugging Face Transformers vs. OpenAI GPT-4 API

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WHAT WE WERE TRYING TO FIND OUT

Our team wanted to answer a practical AI engineering question: if we were adding an AI feature to a real application, when would it make more sense to run a Hugging Face model, and when would it make more sense to call OpenAI through an API? We compared the two approaches using usability, performance, support, scalability, control/privacy, and cost so the report stayed focused on decisions a development team would actually make.

WHAT OUR TEAM ACTUALLY DID

We divided the work across research, reporting, presentation design, repository setup, and team coordination. Brandon organized the PowerPoint and demo work, Elijah led report cleanup and submission organization, Justin helped gather and verify research, Jonah edited the slide wording and checked report-slide alignment, Osemudia helped with roster and project review, and Samantha helped clarify submission/team participation so the correct group materials could be finalized. For the demo, the Hugging Face side used a pretrained sentiment-analysis pipeline, while the OpenAI side used an explicit sentiment prompt sent through an API.

WHAT WE FOUND

Area	Hugging Face	OpenAI API
Usability	More setup; model/environment chosen by developer	Quicker start; app sends requests to hosted model
Performance	Depends on chosen model and hardware	Hosted model; depends on network/API access
Support	Open-source community and documentation	Official platform documentation/service support
Scalability	Team manages serving and infrastructure	Provider manages model hosting
Control / Privacy	More control over model files and deployment	Less model-level control; external API call
Cost	Hardware, hosting, setup, and maintenance	Usage-based API cost

REAL-WORLD EVIDENCE WE USED

JPMorganChase AI Research lists foundation models for the financial domain and multimodal document processing and analysis as research focus areas, which shows how language-model systems can support document-heavy financial work. OpenAI's Duolingo case study explains that GPT-4 was used in Duolingo Max for Role Play and Explain My Answer, giving learners AI conversation practice and feedback without Duolingo building the foundation model from scratch.

WHAT WE LEARNED

The main lesson was that Hugging Face and OpenAI are not just two versions of the same tool. Hugging Face gives the team more control and more responsibility for the model workflow. OpenAI removes much of the infrastructure work and lets the team focus more on the application and prompting. Our final choice would depend on the project data, hardware, budget, timeline, and the team's technical experience.

SOURCES

Hugging Face, "Transformers Documentation" - <https://huggingface.co/docs/transformers/index>

OpenAI, "Developer Quickstart" - <https://platform.openai.com/docs/quickstart>

JPMorganChase, "Artificial Intelligence Research" - <https://www.jpmorganchase.com/about/technology/research/ai>

OpenAI, "Duolingo: Filling crucial language learning gaps" - <https://openai.com/index/duolingo/>